

Conformalized Time Series Anomaly Thresholding with Latent Space Features

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Abstract

Time series anomaly detection is an important task applicable to a wide range of fields. Many methods have been proposed to address this task, but evaluation often suffers leakage from flawed thresholding methods. In this paper we explore leveraging latent space features in a conformal framework to automatically determine alarm thresholds without the risk of leakage and reducing false alarm rates. We show that our method is able to match or surpass the performance of a popular problematic thresholding method while using only information from the calibration set.

Keywords: Time series, anomaly detection, conformal prediction

1. Introduction

Anomaly detection is the general task of identifying outliers within a set of data, which is widely applicable within many domains. In this paper we focus on time series anomaly detection, i.e., the task of identifying unusual points or patterns within data ordered over time. Many real world systems now generate continuous streams of time series data and anomalies within these can represent critical events such as cyberattacks or system faults (Mathur and Tippenhauer, 2016; Hundman et al., 2018). The increase in data availability combined with the natural scarcity of anomalies and laborious annotation process has given rise to unsupervised approaches. These approaches train models based on behavior during normal operation and the predictions represent the expected outcome. The difference between the expectation and the actual observed behavior is used to indicate the presence of anomalies (Garg et al., 2022).

As in many other fields, deep learning methods have seen increased popularity in recent years. Yet recent studies have highlighted the prevalence of data leakage within deep learning literature, leading to overoptimistic reporting and poor real performance (Kapoor and Narayanan, 2023). A major area of leakage within time series anomaly detection is thresholding. To facilitate the use of standard classification metrics of precision, recall, and F1, an alarm threshold must be set to discretize the continuous anomaly scores into binary normal or anomalous classes. A popular thresholding approach used for evaluating classification performance is simply choosing a static threshold that results in the best scores overall, a clear case of leakage. Metrics which measure performance across multiple thresholds such as ROC and VUS (Boniol et al., 2025) have been used to provide a more comprehensive view of model performance, but this sidesteps the question of when should an alarm be raised.

Conformal prediction is a post-hoc framework for distribution free uncertainty quantification which constructs prediction sets with rigorous coverage guarantees (Vovk et al., 2022). With a user defined significance level ϵ , the coverage guarantee refers to the probability of the true value being in the prediction set being greater than or equal to $1 - \epsilon$. We can use conformal prediction for thresholding by reformulating time series anomaly detection from a binary normal/anomalous classification task to a problem with a single normal class, with the task to accept or reject an observation from the normal class. The prediction set is now the set of normal values, and rejections can be considered to be anomalies. This has the additional benefit of limiting the false alarm rate to no more than ϵ . Conformal prediction has been applied along these lines to anomaly detection in previous work (Laxhammar, 2014; Ennadir et al., 2023).

Crucially, conformal prediction assumes that the data is exchangeable, or that each data point is equally likely to occur at any position, to guarantee validity. The temporal dependencies within time series data naturally violates this assumption. Several approaches have been developed to address this limitation in time series data (Stocker et al., 2025). Previous work in time series anomaly detection have employed weighting the calibration set based on relevance to the observation, updating the conformal framework with new observations, and dynamically adjusting the miscoverage level using quantile adjustment (Gil et al., 2026; Yuan et al., 2026). However, these methods train additional black box models to perform reweighting or introduce the possibility of leakage when updating the framework with information from the test dataset.

We propose an alternative method for performing conformal prediction for time series anomaly detection by leveraging the latent space features that are already being learned by models for reweighting and adjusting miscoverage levels. This method does not require additional models to be trained and avoids leakage from updating the framework.

Section 2 gives a brief overview of previous work in anomaly detection and conformal time series methods. Section 3 introduces our proposed method and the results are presented in Section 4. Finally, we summarize our main conclusions and outline directions for future research in Section 5.

2. Previous Work

2.1. Conformal Prediction

Conformal prediction is a powerful distribution free framework for uncertainty quantification (Vovk et al., 2022). Starting with a collection of objects and labels, $\{(X_1, Y_1), \dots, (X_n, Y_n)\}$, where $X_i \in \mathcal{X}$ for some object space $\mathcal{X}$, and $Y_i \in \mathcal{Y}$ for some label space $\mathcal{Y}$, and a test object X_{n+1} , the goal of conformal prediction is to construct prediction sets $C(X_{n+1})$ such that the $\mathbb{P}(Y_{n+1} \in C(X_{n+1})) \geq 1 - \epsilon$ for a given miscoverage level ϵ , and where Y_{n+1} is the (unknown) label for the test object. This set is constructed using nonconformity scores which typically measure an underlying model’s prediction error. Nonconformity scores should be large when model’s performance is poor and vice versa.

Split Conformal Prediction is a common approach to constructing these prediction sets proposed by Papadopoulos et al. (2002). First, the data is partitioned into three disjoint datasets; training, calibration, and test. The model is fit on the training dataset and the

nonconformity score is calculated for every point in the calibration set, creating S_{cal} . To achieve the target coverage of $1 - \epsilon$, we compute a threshold $q_{1-\epsilon}$ by:

$$q_{1-\epsilon} = \text{quantile}(S_{cal}, 1 - \epsilon) \quad (1)$$

The prediction set for a test object X consists of all labels in $Y \in \mathcal{Y}$ such that the nonconformity score for (X, Y) does not exceed this threshold.

2.1.1. UNSUPERVISED TIME SERIES ANOMALY DETECTION

Due to the abundance of available normal data and the relative rarity of labeled anomalous data, time series anomaly detection is often approached as an unsupervised task. Models are trained to represent the underlying distribution of normal behavior, and deviations from the predicted output and observed output are used as the anomaly score. Of particular interest for us are the reconstruction family of models as defined by Darban et al. (2024). These models use the strategy of compressing the input vector into a lower dimensional space and then reconstructing the input from the resulting latent vector. Principal Component Analysis (PCA) can be used in this manner for anomaly detection by comparing the result of decomposing and then reconstructing an observation from its constituent eigenvectors (Lakhina et al., 2004).

As in many other fields, machine learning methods have seen increased popularity for time series anomaly detection due to the ability to model nonlinear relationships. Autoencoders (AEs) use feed forward networks to directly encode and decode sequences from a lower dimensional latent space (Sakurada and Yairi, 2014). Variational Auto Encoders (VAEs) improve the representational power of the latent space by encoding the sequence into parameters of a distribution in the latent space and the reconstruction is then based on sampling from the latent space (Sakurada and Yairi, 2014). Park et al. (2018) propose LSTM-VAE which replaces the feed forward networks in VAEs with LSTMs to better capture temporal relationships.

2.1.2. CONFORMALIZED TIME SERIES

The key theoretical property which allows conformal prediction to provide guaranteed coverage (or validity) is the exchangeability assumption. A sequence is considered to be exchangeable if the elements are equally likely to occur in any permutation. Without this property, it is difficult to establish validity. Stocker et al. (2025) describe several families of methodologies developed to handle this challenge for time series analysis.

1. **Reweighting.** Weight the scores of the calibration set based on their relevance to the current test post.
2. **Refreshing.** Update the calibration set by adding new observations and removing old irrelevant ones.
3. **Adaptive Coverage.** Dynamically update allowed miscoverage level ϵ . Also known as quantile adjustment.

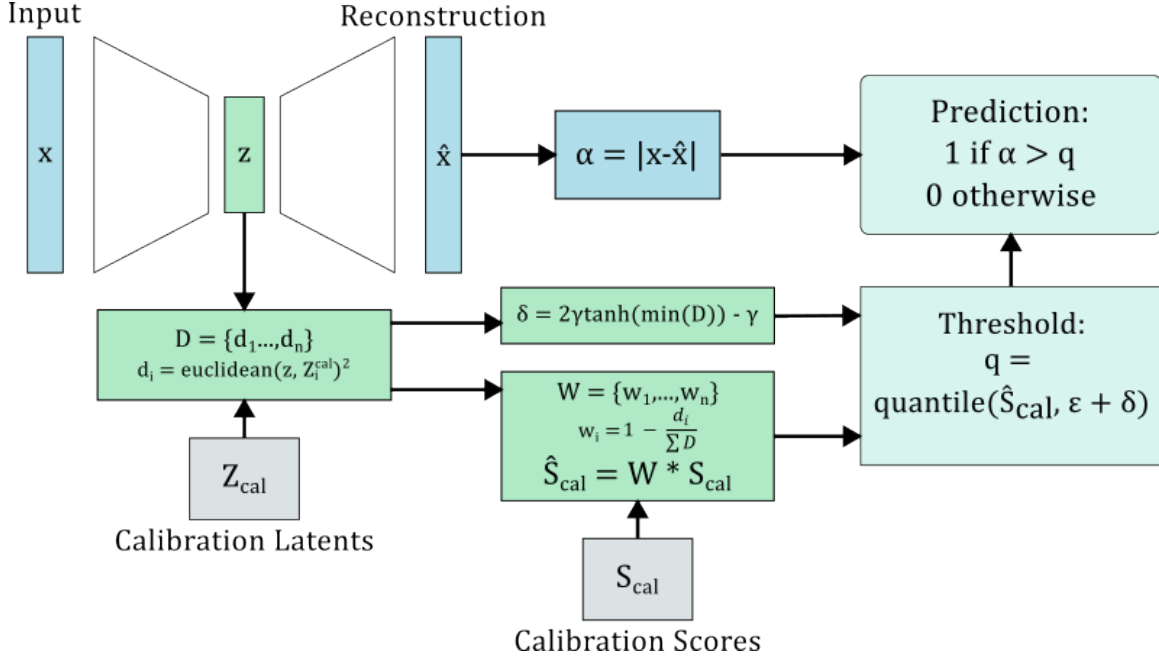


Figure 1: An overview of the proposed method. Latent variables are first extracted from the reconstruction model. Then, the square euclidean distance is calculated between the latent vectors of the observation and the calibration set. These distances are used to calculate calibration weights and the quantile adjustment parameter. The final threshold is determined by taking the quantile of the weighted calibration set with the adjusted ϵ .

4. **Blocking.** Redefine the object of randomization by considering blocks of data exchangeable rather than individual data points.

Previous work in conformal time series anomaly detection have used a combination of these methodologies. Yuan et al. (2026) propose a general method using a novel rejection mechanism to perform refreshing and employ adaptive coverage. Gil et al. (2026) focus on foundational time series models and trains an attention mechanism for reweighting and refreshing.

3. Methods

We propose a method utilizing latent space vectors that are already being generated by reconstruction anomaly detection methods in a conformal prediction framework. Typically, latent spaces have fewer dimensions than the original features and similar inputs map to similar latent space vectors. These two properties make latent space vectors useful for determining similarity while being more computationally tractable than directly using vectors in the original feature space. We use these vectors as part of the reweighting and quantile adjustment strategies and we evaluate thresholding performance across multiple models on a synthetic dataset. An overview of the proposed method can be seen in Figure 1.

3.1. Conformal Prediction

In unsupervised time series anomaly detection, the difference between the model’s input and reconstruction is used as the anomaly score. This score measures irregularity of an observation, therefore in principle the anomaly score is also a suitable nonconformity score. Given observation X and corresponding reconstruction $\hat{X}$, we use the absolute residuals as the nonconformity score:

$$\alpha = \|X - \hat{X}\| \quad (2)$$

The calibration non-conformity scores are collected in S_{cal} . While generating S_{cal} , the latent vectors of the calibration set are extracted to form $Z_{cal} = \{z_1, \dots, z_n\}$.

Next, for any new observation X and its corresponding latent vector z , we calculate the square Euclidean distances $D = \{d_1, \dots, d_n\}$ between z and each vector $z_i \in Z_{cal}$.

3.1.1. REWEIGHTING

We extend the work by [Gil et al. \(2026\)](#) on reweighting the calibration set. Instead of training a new model to learn weights, the calibration set is reweighted relative to a test instance based on latent space distances. Since lower distances indicate higher similarity, the calibration weights $W = (w_1, w_2, \dots, w_n)$ invert the distance measures where:

$$w_i = 1 - \frac{d_i}{\sum_{j=1, \dots, n} d_j} \quad (3)$$

As shown by [Stocker et al. \(2025\)](#) in Theorem B.0.1, the coverage gap of weighting the calibration set is bounded. If larger weights are correctly assigned to calibration points that are similar to the test point, the bound is small.

3.1.2. QUANTILE ADJUSTMENT

We introduce an adjustment parameter δ to adjust the significance level so that the target miscoverage level is higher when the new observation is dissimilar from the calibration set and lower when it is similar.

The tanh function was chosen to map distances to δ . Its properties of passing through the origin and being bounded by one allow for easy control of the maximum possible adjustment with a user defined scaling factor $\gamma \in (0, \epsilon)$. δ is then calculated by:

$$\delta = 2\gamma \tanh(\min(D)) - \gamma \quad (4)$$

When $\gamma = \epsilon$, the range of $\epsilon + \delta$ is then $[0, 2\epsilon]$. [Zhen Lin \(2022\)](#) show that the adjusted target coverage is bounded by the expected value of the adjustment parameter in Theorem 4.1. In our case, the possible adjustment range is symmetric around 0 and bounded by $[-\gamma, \gamma]$. Assuming that the calibration set is representative, the expected value of the adjustment level is near 0 or negative.

dim	wave	freq	coef	offset	noise
0	sin	0.04	1.5	0	0.05
1	cos	0.04	2.5	0	0.05
2	sin	0.04	1.5	0	0.1
3	cos	0.04	2.5	0	0.1
4	sin	0.04	1.5	-2.0	0.05

Table 1: Parameters used to generate the synthetic train dataset

3.1.3. THRESHOLDING

The alarm threshold q is finally calculated by:

$$q = \text{quantile}(WS_{cal}, 1 - (\epsilon + \delta)) \quad (5)$$

An observation will be labeled anomalous if its nonconformity score α is greater than the alarm threshold.

3.2. Dataset

The performance of this approach is tested on synthetic data. We employ the synthetic multivariate dataset generator proposed by [Lai et al. \(2021\)](#) based on noisy sine and cosine waves. We generate a five dimensional multivariate time series of length 500 as the train dataset. The parameters used are described in Table 1.

The last 10% of the train dataset is withheld as the calibration dataset. Test datasets are created by sampling new datasets using the same parameters as in Table 1 and injecting them with following types of anomalies:

1. **Point Global.** Individual instances outside of the normal range of the series.
2. **Point Context.** Individual instances that exist within the normal range of the series, but are contextually anomalous.
3. **Shapelet.** Subsequences with perturbed basic shapelets, for example altered waveforms.
4. **Seasonal.** Subsequences where the shapelets and trend are preserved but the seasonalities are anomalous, for example a shift in frequency.

We create one test dataset for each type of anomaly using the default parameters given by [Lai et al. \(2021\)](#) for a total of four test datasets. Examples of each type of anomaly can be seen in Figure 2. After generation, the train, calibration, and test set are windowed using sliding windows of size 12.

3.3. Underlying Models

This project examines four reconstruction models which generate latent space vectors that are compatible with our proposed method; Principal Component Analysis (PCA), Autoen-

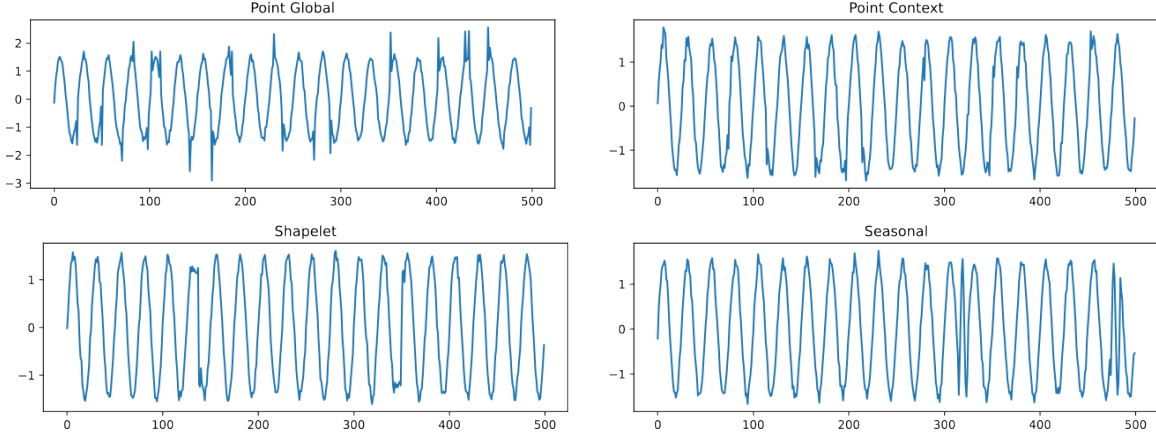


Figure 2: Examples of each type of anomaly generated for this project.

coder (AE), Variational Autoencoder (VAE), and LSTM-Variational Autoencoder (LSTM-VAE) (Lakhina et al., 2004; Sakurada and Yairi, 2014; An and Cho, 2015; Park et al., 2018).

3.4. Experimental Setup

The proposed conformal method was evaluated using a miscoverage level at $\epsilon = 0.05$. Distances are calculated efficiently using the `cdist` function in the SciPy package (Virtanen et al., 2020). We use $\gamma = 0.05$, which sets the possible range of the adjustment parameter δ to be between $[-0.05, 0.05]$.

The PCA model was fit using the default settings in scikit-learn (Pedregosa et al., 2011). Deep learning models share architectural hyperparameters of hidden dim = 16 and latent dim = 3. Each model was trained using the Adam optimizer with default parameters for 100 epochs.

4. Results

This section presents our preliminary results for automatic thresholding by using the latent space in conformal prediction. To assess thresholding performance, we report the widely used F1 score to measure overall classification quality and the False Positive Rate (FPR). The false positive rate represents the rate at which the normal class is falsely rejected, which is equivalent to the miscoverage rate specified by ϵ . The objective is then to maximize the F1 score while keeping the FPR close to or below ϵ .

$$Precision = \frac{TP}{TP + FP} \quad (6)$$

$$Recall = \frac{TP}{TP + FN} \quad (7)$$

$$F1 = 2 * \frac{Precision * Recall}{Precision + Recall} \quad (8)$$

Table 2: Thresholding results on point-global anomalies.

	PCA		AE		VAE		LSTM-VAE	
	F1↑	FPR↓	F1↑	FPR↓	F1↑	FPR↓	F1↑	FPR↓
Best-F1	0.554	0.020	0.933	0.074	0.785	0.107	0.726	0.125
Basic	0.573	0.068	0.877	0.101	0.760	0.058	0.753	0.080
Weighted	0.573	0.070	0.873	0.104	0.758	0.059	0.752	0.080
Adjusted	0.590	0.023	0.934	0.040	0.796	0.043	0.705	0.053
Combination	0.590	0.023	0.934	0.043	0.796	0.045	0.705	0.053

Table 3: Thresholding results on point-context anomalies.

	PCA		AE		VAE		LSTM-VAE	
	F1↑	FPR↓	F1↑	FPR↓	F1↑	FPR↓	F1↑	FPR↓
Best-F1	0.472	0.150	0.703	0.088	0.336	0.294	0.337	0.343
Basic	0.478	0.077	0.712	0.062	0.339	0.083	0.269	0.052
Weighted	0.478	0.077	0.709	0.064	0.339	0.088	0.272	0.054
Adjusted	0.464	0.047	0.735	0.023	0.280	0.034	0.224	0.047
Combination	0.464	0.047	0.735	0.023	0.292	0.036	0.228	0.049

$$FPR = \frac{FP}{FP + TN} \quad (9)$$

Where TP is true positives, FP is false positives, FN are false negatives, and TN are true negatives. For each combination of base model and type of anomaly, we evaluate five thresholding methods.

- **Best-F1.** A post-hoc method which selects a static threshold that results in the best overall F1 score.
- **Basic.** The basic split conformal prediction algorithm with no additional changes.
- **Weighted.** Conformal prediction using only the proposed weighting method.
- **Adjusted.** Conformal prediction using only the proposed quantile adjustment method.
- **Combination** Conformal prediction using both weighting and quantile adjustment.

Best-F1 and basic are intended to be used as baseline comparisons. We would like to note that while Best-F1 remains a problematic measurement of model performance due to leakage, it is included to illustrate the best case scenario for a static anomaly score threshold. Weighted and adjusted show the individual performance of each part of the proposed method.

Table 4: Thresholding results on shapelet anomalies.

	PCA		AE		VAE		LSTM-VAE	
	F1↑	FPR↓	F1↑	FPR↓	F1↑	FPR↓	F1↑	FPR↓
Best-F1	0.462	0.160	0.631	0.154	0.340	0.289	0.329	0.322
Basic	0.484	0.076	0.712	0.061	0.328	0.087	0.308	0.068
Weighted	0.484	0.076	0.708	0.064	0.332	0.090	0.310	0.071
Adjusted	0.513	0.047	0.750	0.022	0.324	0.032	0.300	0.051
Combination	0.513	0.047	0.750	0.022	0.323	0.036	0.302	0.053

Table 5: Thresholding results on seasonal anomalies.

	PCA		AE		VAE		LSTM-VAE	
	F1↑	FPR↓	F1↑	FPR↓	F1↑	FPR↓	F1↑	FPR↓
Best-F1	0.431	0.267	0.792	0.063	0.305	0.287	0.294	0.402
Basic	0.474	0.075	0.794	0.066	0.323	0.084	0.240	0.055
Weighted	0.474	0.075	0.790	0.068	0.324	0.091	0.243	0.058
Adjusted	0.484	0.051	0.838	0.025	0.286	0.044	0.202	0.051
Combination	0.484	0.051	0.838	0.025	0.289	0.045	0.207	0.052

Tables 2, 3, 4, and 5 show results on the point-global, point-context, shapelet, and seasonal anomalies respectively. We would like to emphasize that the goal is not to find the best performing classifier, but to examine the performance of thresholding methods.

In Table 2, the proposed methods perform very well on the point-global anomalies. The FPR is successfully reduced from that of Best-F1 (except in the case of PCA where it is already below the miscoverage threshold) while the F1 scores remain similar or better.

In Tables 3, 4, and 5 we see a sharp decline in model classification performance. These anomalies are more challenging than point-global as they more closely resemble the normal dataset. Poor best-F1 performance indicates that the models are not producing well separated anomaly scores. Nonetheless, the proposed thresholding method remains strong for models that perform well; PCA and AE. As seen in point-global, our method is able to match or exceed Best-F1 for F1 score while reducing the number of false positives. VAE and LSTM-VAE struggle more on these datasets, but we still see comparable or slightly worse thresholding performance combined with a reduction in FPR.

In isolation, weighting appears to be a better thresholding strategy for models with latent space regularization (VAE, LSTM). For models which only perform dimensionality reduction (PCA, AE) adjustment tends to perform better. This could indicate that the more meaningful distances in a regularized latent space have an positive impact on weighting performance, supporting our theory that the latent space can be used to successfully identify relevant windows in the calibration set. However, the overall effect of weighting is often relatively small when compared to the basic split conformal method. In combina-

tion, adjustment consistently has a far greater influence on thresholding performance than weighting whether positive or negative.

Interestingly, our proposed conformal methods are often able to match or even exceed the performance of Best-F1 despite only using information from the calibration set. Empirically, the statistical guarantee of a FPR less than 0.05 holds in most cases for our adjustment method. Due to the influence of the adjusted method on the combined method’s performance, the guarantee also generally holds for the combined method. While the weighting method results in lower FPRs than Best-F1, it is often close to or higher than the target rate of 0.05.

5. Concluding Remarks

This paper addresses the challenge of determining anomaly thresholds for time series anomaly detection without leakage. We propose accomplishing this through a conformal framework that leverages the distance between latent vectors produced by reconstruction methods for weighting and quantile adjustment.

These initial results are a promising indicator that this is an effective method for thresholding anomaly scores, being able to match or even exceed problematic methods which select thresholds that optimize classification metrics while reducing the number of false alarms.

However, we should note that these results are on a controlled synthetic dataset. Future work should include exploration into real world datasets and methods to handle data distribution shifts. Additionally, the weighting method gets close to but does not achieve the target miscoverage level. [Barber et al. \(2023\)](#) observe that overly low weights lead to wider prediction intervals, which may indicate that the weighting algorithm needs further adjustment. Since δ is a mapping between a distance to a scalar value between $-\epsilon$ and a user defined maximum, the function used for its construction is flexible. Deeper investigation into alternative functions, such as sigmoid and bounded exponential, could be carried out.

In deep learning for time series anomaly detection, the task assumes that the train dataset is representative of the test dataset. How covariate shifts should be labeled and whether data leakage becomes acceptable in such circumstances should be considered carefully. Other areas to investigate are the performance of the distance calculations, the effect of increasing latent space and window sizes, and the impact of different nonconformity scores. Another avenue of exploration is the effects of preprocessing techniques for time series data on conformal performance.

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